



**AN INTERACTIVE DIGITAL BANKING PLATFORM FOR
ENHANCING FINANCIAL LITERACY AMONG CHILDREN AND
TEENEGERS**

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DECLARATION

We, the undersigned, hereby declare that the report titled “An Interactive Digital Banking Simulation Platform for Children and Teenagers” is our original work submitted in partial fulfilment of the requirements for the award of the Bachelor of Science in Software Engineering degree at Mbarara University of Science and Technology. We further confirm that this report has not been submitted for any other degree and that all sources of information have been duly acknowledged.

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ABSTRACT

Financial literacy is a critical life skill that enables individuals to make informed and effective financial decisions. In Uganda, however, financial literacy levels among children and youth remain low, largely due to limited exposure to practical financial education in schools and homes. This project proposes the design and development of an interactive digital banking simulation platform aimed at improving financial literacy among children aged 5–17 years. The system will provide a gamified learning environment where learners can simulate real-world financial activities such as earning, saving, spending, and investing within a safe, virtual ecosystem. Using intuitive graphics and age-appropriate language, the platform will integrate educational modules, interactive challenges, and parental supervision features to ensure both engagement and learning outcomes. The simulation will also reflect Uganda’s financial context, incorporating local banking practices, mobile money systems, and currency. The proposed solution seeks to foster financial responsibility, enhance decision-making skills, and build early awareness of financial systems among Ugandan children. The project will employ a user-centered design approach, combining educational theory, software development methodologies, and gamification principles. Evaluation will be conducted through usability testing and feedback from learners, educators, and parents. Ultimately, the platform aims to contribute to Uganda’s broader financial inclusion goals by equipping the next generation with essential financial knowledge and digital competencies.

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Table 1: List of Acronyms

Acronym	Meaning
API	Application Programming Interface
AT	Automated Teller Machine
DBMS	Database Management System
FSDU	Financial Sector Deepening Uganda
ICT	Information and Communication Technology
IFL	Interactive Monetary Fund
IMF	International Monetary Fund
MoFPED	Ministry of Finance, Planning and Economic Development
NFIS	National Financial Inclusion Strategy
OECD	Organization for Economic Co-operation and Development
SDG	Sustainable Development Goal
UBOS	Uganda Bureau of Statistics
UI	User Interface
UX	User Experience
UNICEF	United Nations International Children's Emergency Fund
JS	JavaScript
OTP	One-Time Password
PIN	Personal Identification Number
RBAC	Role-Based Access Control
AES	Advanced Encryption Standard
TLS	Transport Layer Security
LAN	Local Area Network
SMS	Short Message Service
JWT	JSON Web Token
DB	Database
API Gateway	Interface managing external service communication
CRUD	Create, Read, Update, Delete
UI/UX	User Interface / User Experience Design
REST	Representational State Transfer
JSON	JavaScript Object Notation
HTTPS	HyperText Transfer Protocol Secure

1 CHAPTER ONE: INTRODUCTION

Financial literacy is the ability to understand and effectively apply financial knowledge and skills to make informed decisions about managing money. It involves budgeting, saving, investing, and responsible borrowing core competencies that influence an individual's financial well-being and stability. In a world where financial systems are increasingly digital and complex, financial literacy has become a fundamental life skill, essential for achieving personal independence and contributing to national economic growth. However, despite its importance, many people, especially in developing countries like Uganda, lack the basic financial knowledge necessary to navigate daily financial decisions. This challenge is even greater among children and teenagers, who often reach adulthood without understanding how to manage income, control spending, or plan for the future. The absence of early financial education means that young people grow up with weak saving habits, poor budgeting discipline, and limited awareness of how financial systems work. Consequently, they become vulnerable to a lot of debt, financial exploitation, and poverty traps that could have been avoided through early guidance. In Uganda, where over 75 % of the population is below the age of 30 while 50.5% of the population is under 18 [16]. The lack of financial literacy among youth represents a significant economic and social challenge. Schools rarely include financial management in their curricula, and parents often lack both the tools and the confidence to teach money management at home. While financial inclusion has expanded through mobile money and digital banking, these services largely target adults, leaving children and teenagers excluded from practical financial experiences. The result is a generation that interacts with money daily but lacks the knowledge and skills to manage it wisely. This gap calls for innovative, interactive, and age-appropriate solutions that make financial learning both engaging and practical. Therefore, this study proposes the development of an interactive digital banking simulation platform designed to teach children and teenagers the principles of saving, budgeting, and responsible spending in a safe, supervised, and enjoyable environment. By blending technology with education, the platform aims to build a financially literate generation capable of making informed decisions and contributing meaningfully to Uganda's future economy.

1.1 Background

Globally, financial literacy has become a critical skill in navigating today's complex financial environment. As economies expand and financial products become more complex, individuals need the knowledge and skills to make informed financial decisions. According to the World Bank (2019), nearly two-thirds of adults worldwide lack basic financial understanding, making poor financial decisions that lead to high debt levels, low savings, and financial insecurity. Financial literacy is no longer optional it is essential for ensuring personal and national economic stability in a world driven by financial systems and digital transactions. In the United States, despite being one of the

world's largest economies, financial illiteracy remains a significant problem. A survey by the National Financial Educators Council (2022) revealed that financial illiteracy costs Americans more than USD 400 billion annually due to poor decision-making, high-interest debt, and inadequate saving habits. Many young adults enter the workforce with little understanding of how to manage credit, invest, or budget effectively, despite the country's advanced banking and education systems. In Europe, research shows a similar trend. The Organization for Economic Cooperation and Development (OECD, 2021) found that even in high-income European countries, a large percentage of citizens lack sufficient financial knowledge to manage their finances effectively. Although many European nations have integrated financial literacy into their school curricula, challenges persist in applying theoretical knowledge to real-life situations. For example, in countries such as Spain and Italy, youth often struggle with budgeting and debt management despite national awareness programs. In Asia, rapid economic growth and urbanization have increased access to financial services, but not necessarily to financial knowledge. Studies in China and India indicate that a large portion of the population, especially in rural areas, remains financially illiterate. Xu and Zia (2012) observed that many Asian youths lack basic knowledge about saving and credit management, making them vulnerable to financial mismanagement. Governments in countries like Singapore and Japan have introduced financial education initiatives at the school level, but the gap between financial access and financial understanding still persists. In Africa, financial literacy remains one of the continent's major development challenges. The rise of mobile money services has improved financial inclusion but has not necessarily increased financial knowledge. According to the OECD (2020), many African adults and youths use financial services without understanding the implications of interest rates, budgeting, or saving. In countries such as Kenya, Ghana, and Nigeria, governments and NGOs have implemented financial education programs, yet progress has been slow due to limited resources, low digital literacy, and a lack of structured financial education within schools. In Uganda, the situation mirrors that of many other African countries. Despite the expansion of mobile money platforms, SACCOs, and microfinance institutions, financial literacy levels remain low, especially among the youth. Kasozi and Makina (2021) report that only about one third of Ugandans demonstrate basic financial knowledge. The National Financial Inclusion Strategy (2017–2022) identified financial literacy as one of the five pillars for improving livelihoods, but implementation especially in schools remains limited. With over 75% of the population below the age of 30 [16], Uganda faces the challenge of preparing a young generation for financial independence without sufficient financial education. The lack of structured, practical financial learning opportunities means that many youths enter adulthood without essential skills in budgeting, saving, and responsible spending. This global-to-local perspective highlights a universal gap in financial literacy and emphasizes the need for innovative, interactive, and technology-based approaches to equip young Ugandans with essential financial management skills early in life

1.2 Problem Statement

Financial literacy among children and teenagers in Uganda remains critically low, largely due to the lack of structured financial education in schools, limited parental involvement, and the absence of interactive, age-appropriate digital tools that make learning about money practical and engaging. This issue predominantly affects young people aged 5–17, who form a significant portion of Uganda’s youthful population, with over 75% of citizens under the age of 30[16] . As a result, many youths enter adulthood without the necessary skills to budget, save, or manage money responsibly, leading to poor saving habits, financial vulnerability, and reduced economic participation. If this challenge is not addressed, Uganda risks raising a generation that remains financially dependent and ill-equipped to contribute meaningfully to the country’s financial inclusion and economic growth efforts. Therefore, this study proposes the development of an interactive, parent-supervised digital banking simulation platform designed to enhance financial literacy among children and teenagers through engaging, practical simulations.

1.3 General Objective

To improve on the financial literacy among children and teenagers in Uganda by developing an interactive digital banking simulation platform.

1.4 Specific Objectives

- i. To conduct a comprehensive literature review on existing knowledge and research gaps related to the financial literacy among teenagers and young children.
- ii. To design and develop a web and mobile application that is interactive, age-appropriate digital banking simulation platform that teaches budgeting, saving, and responsible spending.
- iii. To test and validate the effectiveness of the application in improving financial knowledge, practical money management skills, and confidence among young users.

1.5 Research Questions

- i. How can an interactive digital banking simulation platform be designed and evaluated to effectively enhance financial literacy among children and teenagers in Uganda?

- ii. What does existing literature reveal about the current state of financial literacy among children and teenagers, and what gaps exist in existing educational approaches?
- iii. What key features and design elements are required to develop an interactive and age appropriate digital banking simulation platform that effectively teaches budgeting, saving, and responsible spending?

1.6 Project Significance

This project is significant because it introduces a structured and controlled digital banking system designed to promote financial literacy among children while maintaining parental oversight. The system provides a safe environment where children can learn how to manage money, set savings goals, and make financial decisions, while parents retain authority over critical actions such as withdrawals and spending. This balance helps children develop responsible financial habits at an early stage. The system also demonstrates the practical application of modern software engineering technologies in building secure and scalable financial solutions. By integrating a cross-platform frontend, a Type Scriptbased backend, and a structured database system, the project highlights how real-world banking functionalities can be implemented using contemporary development tools and frameworks.

1.6.1 Value Proposition

The proposed system will provide children and teenagers with an engaging platform to learn essential money management skills through interactive simulations. It will also empowers parents and educators to monitor progress and guide financial learning. By integrating gaming elements and real-world scenarios, the system will offer a unique, fun, and practical way to build financial confidence at an early age.

1.6.2 Innovation

This project will introduce an innovative digital banking simulation platform designed specifically for children and teenagers. It will combine parental supervision, and interactive simulations to provide hands-on learning experiences. Unlike existing financial literacy initiatives, the platform will allow learners to virtually earn, save, spend, and invest money in a safe environment— promoting experiential learning rather than passive instruction.

1.6.3 Impact

The project is expected to improve the financial literacy levels of Uganda's youth population, fostering a generation that is more financially responsible and independent. By cultivating early saving habits and responsible spending behavior, the system contributes to reducing debt vulnerability, enhancing household savings, and supporting Uganda's broader financial inclusion goals.

1.6.4 Business Component

The system has the potential to evolve into a sustainable business venture. Potential users include schools, parents, NGOs, and government programs promoting youth empowerment. The business model may include subscription plans for schools, partnerships with financial institutions for sponsorship, and localized versions for different age groups. This ensures longterm sustainability while contributing to national education and financial inclusion efforts.

1.6.5 Expected Outcomes

A fully functional interactive digital banking simulation platform tailored for children and teenagers aged 5–17 in Uganda. Improved financial literacy levels, including knowledge of budgeting, saving, and responsible spending. Enhanced practical money management skills and decision-making confidence among young users. Insights into usability, accessibility, and educational impact, which can inform future financial education initiatives and policy development

1.7 Scope of Study

The study focuses on the design, development, and evaluation of an interactive digital banking simulation platform for children and teenagers aged 5–17 in Uganda. The platform will include features for saving, budgeting, and spending simulations, integrated with parental supervision. The implementation and testing will primarily target selected schools and households in urban centers such as Mbarara. The project's findings will provide insights into improving financial education strategies and inform policymakers, educators, and developers about effective digital literacy tools for young learners.

2 CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature related to financial literacy, digital learning, gamification, and simulation-based education. The purpose is to establish the theoretical and empirical foundations for developing an interactive digital banking simulation platform designed to enhance financial literacy among Ugandan children and teenagers. The review discusses existing financial literacy programs, their effectiveness, and the emerging role of technology and experiential learning in financial education.

2.2 According to objective(i) Financial Literacy and its Importance

Financial literacy is the ability to understand and effectively manage personal finances through budgeting, saving, and responsible spending. According to Johnson and , financial [8]education programs are most effective when tailored to individual needs and behavioral patterns rather than delivered as generic, one-size-fits-all solutions. Their findings highlight the importance of personalized and experiential approaches, particularly for young learners who are still developing foundations in Financing. Studies indicate that financial literacy levels among youth are generally low across both developed and developing countries. In the United States, the National Financial Educators Council (2022) found that many high school and college students lack basic knowledge of budgeting, credit, and saving, resulting in poor financial decisions and high levels of student debt. Similarly, in Europe, research by the OECD (2020) shows that despite the inclusion of financial education in school curricula, young people often struggle to apply financial concepts to real-life situations. In Asia, Xu and[18] observed that children and teenagers in China and India possess limited knowledge about credit, savings, and financial planning, particularly in rural areas. While urban youth have greater access to banking and digital services, understanding of financial principles remains weak, highlighting a gap between access and knowledge. These findings suggest that early, structured financial education is critical to improving youth financial behavior. In Africa, financial literacy among youth is similarly low. According to the[14], despite widespread use of mobile money, many children and teenagers lack understanding of saving, budgeting, and interest implications. In countries such as Kenya, Ghana, and Nigeria, NGOs and government initiatives have introduced youth financial education programs; however, progress has been limited due to insufficient school integration, resource constraints, and low digital literacy. Financial literacy among young people is a major concern. [9] report that only about one-third of Ugandans demonstrate basic financial knowledge. This situation is compounded by limited financial education in schools and low parental involvement in money management guidance. While mobile money platforms have increased financial

inclusion, these services primarily target adults, leaving children and teenagers with limited practical experience in managing money. The National Financial Inclusion Strategy (2017–2022) identifies financial literacy as a key pillar for improving livelihoods. Yet, implementation at the school level remains minimal. As over 75% of Uganda’s population is under 30, this demographic gap presents a significant challenge for national economic development. Young people entering adulthood without budgeting, saving, or spending skills are at higher risk of debt and financial mismanagement, highlighting the urgent need for innovative approaches to youth financial education. In Uganda, low financial literacy among youth remains a major concern due to limited access to formal financial education. Existing programs often focus on adults, leaving children and teenagers without practical tools to learn about money management. This gap emphasizes the need for early, interactive, and accessible financial education models.

2.2.1 Digital Platforms in Financial Education

The rise of digital platforms has created new opportunities for delivering financial literacy education. [10] observed that participants who engaged with financial content through online communities showed significant improvements in both financial knowledge and confidence. Their study demonstrates that digital platforms can make financial learning social, scalable, and engaging, particularly for younger audiences who are already comfortable using mobile and web technologies. This aligns closely with the concept of a digital banking simulation platform, which leverages interactivity and connectivity to promote financial learning. By simulating real banking activities in a supervised and gamified environment, children can gain practical knowledge about saving, spending, and budgeting through active participation rather than passive instruction.

2.2.2 Role of Teachers and Educational Integration

The effectiveness of financial literacy programs also depends on the quality of delivery and teacher preparation. According to [3] and Collins and [2], professional development for educators is vital in equipping them with the tools and confidence to integrate financial education into existing curricula. However, in many developing contexts like Uganda, limited teacher training and resources restrict classroom-based financial education. This challenge supports the introduction of technology-driven tools that can supplement or even replace traditional instruction. A self-guided digital platform can enable children to learn independently, while allowing parents and teachers to monitor progress through built-in supervision features.

2.2.3 Gamification and Experiential Learning in Financial Education

Gamification is the use of game-like elements such as points, badges, levels, and leaderboards. This has been shown to significantly enhance student motivation and learning outcomes [15]. These elements make abstract financial concepts more tangible, turning learning into a fun and competitive experience. Incorporating Social and Emotional Learning (SEL) principles into gamified systems also fosters critical life skills such as decision-making, goal-setting, empathy, and self-awareness [12]. These are directly linked to responsible financial behavior, making gamification a suitable strategy for teaching children about money management. [6]

2.2.4 Experiential Learning Models

Experiential learning involves teaching through direct experience and reflection rather than traditional lectures. Several financial literacy programs have successfully adopted this approach. For instance: My Classroom Economy (MyCE) allows students to participate in a classroom-based 'microeconomy,' earning and spending 'school dollars.' This simulation led to a 6% increase in financial knowledge among learners (Batty et al., 2015). The Family Financial Fun Learning (FFFL) curriculum combines role-playing, discussions, and case studies to reinforce financial concepts. It achieved an 11% increase in financial knowledge among learners (Batty et al., 2015). The Stock Market Game (SMG) engages students in managing a virtual investment portfolio, leading to substantial gains in financial understanding [7]. These examples demonstrate that simulation-based learning enables students to practice realworld financial decision-making in a risk-free environment. This directly supports the core idea of your proposed system — a digital banking simulator that combines education, interaction, and fun to enhance financial literacy [6].

2.2.5 Simulation-Based Financial Literacy Programs

Interactive and experiential approaches to financial literacy have been widely recognized as effective in improving young learners' understanding of money management [17] evaluated the Money Talks: Should I Be Listening? financial education curriculum designed for teenagers, which incorporated hands-on activities, games, and scenario-based exercises. Their study revealed that students who participated in the program demonstrated significant improvement in both financial knowledge and behavior. These insights align closely with the objectives of the present study, which seeks to design an interactive digital banking simulation platform. By engaging children and teenagers in simulated financial activities—such as budgeting, saving, and spending—the proposed system applies the same experiential principles proven effective in Varcoe et al.'s work. This connection supports the idea that gamified and practical learning environments can drive long-term changes in financial attitudes and

habits among youth. Similarly, [1] investigated the impact of a financial literacy simulation known as The Reality Store among middle school students. Their research demonstrated that simulation-based activities significantly enhanced students' understanding of real-world financial decision-making, including budgeting, income management, and prioritization of needs versus wants. Participants reported that the simulation helped them visualize the consequences of financial choices and increased their motivation to plan for future financial responsibilities. The authors concluded that simulation-based financial education provides a realistic, engaging, and effective framework for teaching money management to young learners, even in the absence of specialized teachers. This finding is particularly relevant to the Ugandan context, where structured financial education is limited and many schools lack trained financial literacy instructors. By integrating a digital simulation approach similar to The Reality Store, the proposed platform aims to fill this educational gap. Through supervised digital experiences, students can gain financial competence in a safe, interactive environment that mirrors real-life banking operations. Together, the works of [1] reinforce the importance of experiential and simulation-based learning models in developing effective financial literacy interventions. They provide empirical evidence that interactive digital tools can improve engagement, comprehension, and practical skill application among young learners. These findings form the foundation for adopting simulation and gamification as central design elements in the proposed digital banking platform.

2.2.6 Existing Financial Literacy Solutions for Youth

Several financial products and applications across Africa and Uganda aim to promote financial literacy among children. Studies show that digital gamification improves financial learning outcomes. For instance, [5] demonstrated that FinSOLEkid significantly enhanced children's financial awareness through interactive digital games. Similar simulation programs, such as The Reality [13], have proven effective in teaching decision-making and budgeting through experiential learning.

Application	Region	Strengths	Gaps
Salaam Junior Account	Uganda	Encourages saving and budgeting with parental supervision through a real bank account	Limited to savings; lacks interactive or educational modules
Future Kids Account	Uganda	Promotes saving culture and introduces children to formal banking	Limited to savings; lacks interactive or educational modules

Earlybean	Africa	Allows children to earn money via chores, set savings goals, and track spending under parental guidance	Focuses on budgeting and saving; limited financial simulation or localization for Ugandan currency and mobile-money systems
MoneyAfrica Kids	Africa	Provides gamified financial lessons and quizzes; localized to African youth	Lesson-based learning; lacks immersive banking simulations or feedback mechanisms
FinSOLEkid	Indonesia	Uses a gamified digital app to teach early-childhood financial literacy; proven improvement in financial understanding	Developed for a different cultural context; not localized for African or Ugandan users

Table 2: Comparison of Existing Applications

2.2.7 Research Gap

From the reviewed literature and existing platforms, three key gaps are evident: i. Limited interactive and experiential financial literacy tools for children in Uganda. ii. Insufficient integration of gamification and simulation in early financial education. iii. Lack of localized, culturally relevant digital platforms aligned with Ugandan financial contexts and youth behavior. The proposed interactive digital banking simulation platform addresses these gaps by combining technology, education, and behavioral design to improve financial literacy outcomes among children and teenagers.

2.2.8 Summary

This chapter reviewed existing literature on financial literacy education, highlighting how digital learning, gamification, and experiential simulation can enhance young learners' understanding of money management. Evidence from global and regional studies shows that engaging, technology driven tools are far more effective than traditional classroom methods.

2.3 According to objective(ii) To design and develop an interactive, age-appropriate digital banking simulation platform for teenagers

Financial literacy among teenagers is increasingly recognized as essential for developing responsible financial behavior, yet many young individuals lack practical skills in budgeting, saving, and spending. Traditional classroom-based financial education has

been found to be less effective because it focuses mainly on theoretical knowledge without providing real-life application. As a result, there is a growing emphasis on experiential learning approaches that allow teenagers to actively engage with financial concepts in practical contexts [11]. Recent studies show that web and mobile-based applications provide effective platforms for delivering financial education to young users. These systems enable simulation of real-world financial activities such as managing money, tracking expenses, and setting savings goals, which improves understanding and retention. Interactive digital tools also offer immediate feedback, helping users learn from their decisions and gradually build better financial habits over time [4]. Furthermore, the integration of gamification and user-centered design has been identified as critical in engaging teenagers. Features such as goal-setting, rewards, and intuitive interfaces make financial learning more appealing and accessible. Additionally, systems that incorporate parental supervision mechanisms help balance independence with guidance, reinforcing responsible financial behavior. These findings support the development of interactive, age-appropriate digital banking platforms for effective financial education [6].

2.4 Objective(iii) To test and validate the effectiveness of the application in improving financial knowledge, practical money management skills, and confidence among young users

Evaluating the effectiveness of financial literacy interventions is essential to ensure that digital applications produce measurable improvements in users' knowledge and behavior. Research indicates that financial education programs can positively influence financial knowledge and decision-making, particularly when assessment methods such as pre- and post-tests are used to measure learning outcomes. However, the impact varies depending on how interactive and practical the learning experience is, highlighting the need for well-designed evaluation approaches [11]. Studies further show that simulation-based and experiential learning tools are more effective in improving practical money management skills compared to traditional teaching methods. Applications that allow users to actively engage in tasks such as budgeting, saving, and managing expenses help reinforce realworld financial behaviors. These systems also enable continuous monitoring and feedback, which are critical for validating whether users are developing sustainable financial habits over time [4]. In addition, user confidence is an important indicator of the success of financial literacy applications. Research suggests that increased financial knowledge and repeated interaction with digital financial tools contribute to higher self-efficacy and confidence in managing money. Validation methods such as user surveys, behavioral analysis, and usability testing are commonly used to assess these outcomes. These findings support the use of structured evaluation techniques to determine the effectiveness of interactive financial applications for young users [6].

3 CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter outlines the prototyping methodology that will be employed in the development of the interactive digital banking simulation platform for enhancing financial literacy among children and teenagers in Uganda. The methodology is structured around iterative prototyping stages aimed at refining system functionalities based on user and stakeholder feedback. The development process progresses through quick design, prototype building, testing, feedback collection, refinement, and final implementation. This approach will ensure that the final system is user friendly, educationally effective, and will be aligned with the learning needs of young users.

3.2 Prototyping Methodology Overview

The researchers will use prototyping as the methodology for system development. Prototyping is a software development process that allows developers to create an early version of a system to demonstrate functionality and make necessary refinements before producing the final product. It involves iterative cycles of design, development, testing, and improvement, enabling developers to rapidly adapt to user feedback and evolving requirements.

3.2.1 Why Prototyping Methodology

The Prototyping Methodology will be chosen for this project due to its flexibility and its suitability for developing interactive systems that require continuous user input and evaluation. The proposed digital banking simulation platform targets children and teenagers, meaning usability, engagement, and user experience are critical. Prototyping allows for early visualization of the system, continuous feedback from users (children, parents, and educators), and iterative refinement to ensure that the final platform is effective, safe, and age-appropriate. This approach minimizes the risks of user dissatisfaction, addresses uncertainties in user requirements, and ensures the system delivers a fun and educational experience that promotes financial literacy.

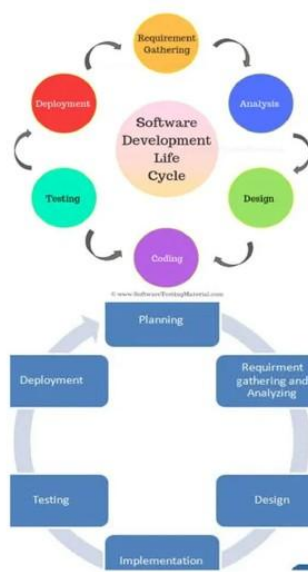


Figure 1: Caption

3.2.2 Prototyping Methodology Illustration

The following are the steps that will be carried out in the proposed project as illustrated below.

3.3 Requirement Gathering

This phase involved collecting detailed insights into the requirements and preferences for the interactive digital banking simulation platform using the following tools and techniques:

3.3.1 Interviews

Interviews were conducted with few key stakeholders including parents, children and financial education experts. These interviews were designed to gather qualitative data on current financial education practices, challenges faced in teaching children about money, and expectations from a digital learning tool. Open-ended questions encouraged participants to share their experiences and opinions on how best children can learn financial concepts.

3.3.2 Questionnaires

To complement the interview data, A structured questionnaire consisting of 20 questions was used to gather user requirements focusing on financial behavior, system features, usability, and security expectations. The questionnaires aimed to collect quantitative information on user needs, preferred features, and levels of financial literacy among children. Respondents rated the importance of various learning modules such as budgeting, saving, and spending, and provided feedback on the kind of gamified elements that would enhance engagement. A sample of the questionnaire is below.

3.3.3 Requirements Gathering Questionnaire

This questionnaire was designed to collect user requirements and understand financial behaviors, system expectations, usability preferences, and security concerns for the proposed Kids Banking System.

Section A: General Information

1. What is your age group?
10–13 14–17 18+
2. Are you a:
Student (Teen) Parent/Guardian
3. Have you ever used a banking or money management app before?
Yes No

Section B: Financial Knowledge and Behavior

4. How would you rate your current understanding of managing money?
Very Low Low High Very High
5. Do you currently track your spending?
Yes No
6. Do you set savings goals?
Yes No
7. What challenges do you face when managing money?

Overspending Lack of knowledge No tracking system Other: _____

Section C: System Features (Requirements)

8. Would you find it useful to have a mobile app that helps you manage money?
Yes No
9. Which features would you like in the system? (Select all that apply)
Account balance tracking
Savings goals
Transaction history
Budget planning
Notifications/reminders
10. Should children require parental approval before withdrawing money?
Yes No
11. How important is it for parents to monitor children's financial activities? Not important Slightly important Important Very important
12. Would you like the system to provide tips on saving and budgeting?
Yes No

Section D: Usability and Design

13. What type of interface do you prefer? Simple
Interactive Game-like
14. How important is ease of use in the application? Not important Important Very important
15. Would you prefer the system to be available on:
Mobile Web Both

Section E: Security and Trust

16. Do you think secure login (password/OTP) is important?

Yes No

17. Would you trust a system that records and tracks all transactions?

Yes No

18. How comfortable are you with sharing financial data in an app?

Not comfortable Neutral Comfortable

Section F: Feedback and Improvement

19. What additional features would you like to see in the system?

20. Do you think this system would help improve financial literacy among children?

Yes No Not sure

em would help improve financial responsibility among teenagers? Yes No Not sure

3.3.4 Observation

Observational methods were employed to understand how children currently interact with digital learning tools and games. This helped to identify patterns in their learning behavior, attention span, and preferences for interactivity and visual design. The observational findings provided contextual data that guided the design of the platform's interface and user experience. Based on these findings, the system requirements were categorized into functional and non-functional requirements.

3.3.5 FUNCTIONAL REQUIREMENTS

3.3.6 NON-FUNCTIONAL REQUIREMENTS

3.4 Quick Design

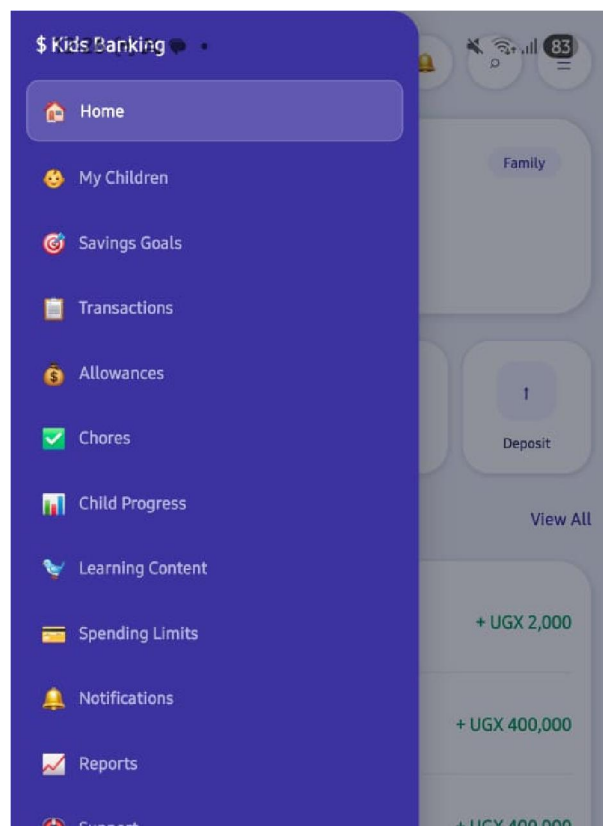
In this phase, the researchers developed an initial conceptual design of the system based on the requirements gathered. The design focused on the core functionalities of the platform such as budgeting, saving, spending simulations, and parental supervision.

Prototyping tools such as Figma were used to create sketches and mockups of the interface. The design prioritized simplicity, colorfulness, and interactivity to appeal to children while remaining educational. The layout was organized to ensure smooth navigation and easy understanding of financial activities.

ID	Requirement
FR1	The system shall allow parent registration with OTP verification (name, phone, email, password).
FR2	The system shall allow parents to create and manage multiple child accounts.
FR3	The system shall store child profile details including age, learning level, savings goals, and restrictions.
FR4	The system shall provide secure authentication (parent: password + OTP; child: PIN/biometric/device login).
FR5	The system shall enforce role-based access control between parent and child users.
FR6	The system shall create and manage a digital wallet for each child with a unique ID.
FR7	The system shall allow parents to fund child accounts and record deposit details (amount, source, timestamp, reference).
FR8	The system shall allow children to view balances and transaction details but restrict modifications.
FR9	The system shall allow children to request withdrawals and require parent approval via OTP/PIN.
FR10	The system shall allow parents to approve, reject, delay, or limit withdrawal requests.
FR11	The system shall support transfers (internal allowed; external requires parent approval).
FR12	The system shall allow children to create and track savings goals and basic budgets.
FR13	The system shall support allowance scheduling, chores assignment, and reward management.
FR14	The system shall provide parental controls, audit logs, reporting, customer support, and optional financial education features.

Table 3: Functional Requirements

ID	Requirement
NFR1	The system shall ensure data security using encryption (AES for data at rest, TLS for data in transit).
NFR2	The system shall securely store passwords and OTPs using hashing and enforce expiry.
NFR3	The system shall implement role-based access control and protect against unauthorized access.
NFR4	The system shall protect user privacy and collect only necessary data with consent.
NFR5	The system shall maintain high availability and uptime for continuous access.
NFR6	The system shall provide fast performance with API response times under 2 seconds.
NFR7	The system shall deliver OTPs within 10 seconds under normal conditions.
NFR8	The system shall support scalability to accommodate growing users and integrations.
NFR9	The system shall ensure reliability by preventing duplicate or inconsistent transactions.
NFR10	The system shall provide a user-friendly interface (simple for children, detailed for parents).
NFR11	The system shall be modular to support easy maintenance and updates.
NFR12	The system shall integrate with external services (SMS, email, payment gateways, mobile money).
NFR13	The system shall maintain auditability of all transactions and administrative actions.



NFR14	The system shall comply with financial regulations, data protection laws, and support backup and disaster recovery mechanisms.
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Table 4: Non-Functional Requirements

3.5 Build Prototype

The developed prototype of the digital banking simulation platform provides several key functionalities aimed at supporting financial learning and controlled money management for teenagers. The application includes a virtual wallet system that allows users to view their balance, simulate deposits, and request withdrawals. This feature helps users understand how money flows in and out of an account without involving real financial risk. The system also includes a budget planning module, which enables users to allocate money into different categories such as savings, spending, and goals. This helps teenagers learn how to manage limited resources and make informed financial decisions. In addition, the application tracks all transactions and provides a transaction history, allowing users to monitor their financial activities and understand their spending patterns over time.

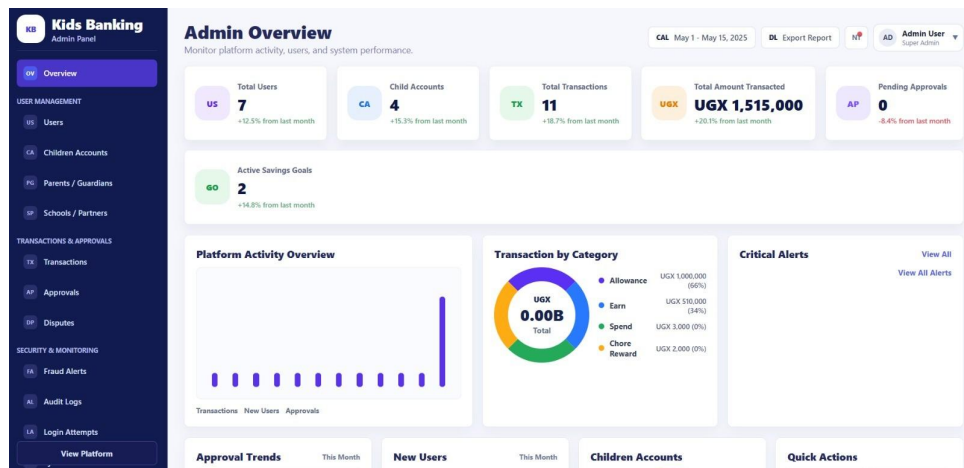


Figure 2: parent dashboard

To improve engagement, the prototype incorporates gamification features such as points, badges, and rewards. These elements motivate users to complete financial tasks like saving regularly or staying within budget. The system also includes a parental supervision dashboard, where parents can monitor activities, approve or reject withdrawal requests, and guide their children's financial behavior. Overall, the prototype demonstrates how digital tools can be used to teach budgeting, saving, and responsible spending in an interactive and controlled environment.

3.6 Test Prototype

Once the prototype was developed, it was tested by a small group of target users, including children, parents, and teachers. The purpose of testing was to evaluate the usability, educational effectiveness, and visual appeal of the platform. User testing sessions were conducted with selected children within Mbarara, where participants interacted with the system under supervision. Data was collected through surveys and observation to determine how easily users shall navigate the platform, understand financial concepts, and stay engaged. Feedback from this was documented and analyzed to identify issues, challenges, and opportunities for improvement.

3.7 Prototype Testing Questionnaire

This questionnaire is designed to evaluate the usability, functionality, and effectiveness of the system.

A: General Information

1. What is your role?
Child Parent Teacher
2. What is your age group?
Below 13 13–17 18+

B: Usability (Ease of Use)

3. Was the application easy to navigate?
Very Difficult Di
4. Were you able to _____
understand how to use the features without help?
Yes No
5. Which part of the system was most difficult to use?

6. How would you rate the overall user interface?
Poor Fair Good Very Good Excellent

C: Functionality

7. Did the system perform all tasks as expected (e.g., saving, requesting withdrawals)?

Yes No

8. Did you experience any errors or system problems?

Yes No

If yes, please specify:

9. Were the features (wallet, budget planner, approval system) useful?

Not useful Slightly useful Useful Very useful

D: Educational Effectiveness

10. Did the application help you understand how to manage money?

Yes No Not sure

11. Did you learn how to save and budget better after using the system?

Yes No

12. How clear were the financial concepts presented in the system?

Not clear Slightly clear Clear Very clear

E: Engagement & Design

13. Did you find the application interesting and engaging?

Not engaging Slightly engaging Engaging Very engaging

14. Did the rewards (points, badges) motivate you to use the system?

Yes No

15. What did you like most about the system?

F: Parental Control (For Parents/Teachers)

16. Was the approval system for withdrawals useful?

Yes No

17. Did the system give you enough control over the child's activities?

Yes No

G: Overall Evaluation

18. How would you rate the overall system?

Poor Fair Good Very Good Excellent

19. Would you recommend this system to others?

Yes No

20. What improvements would you suggest for the system?

3.7.1 Feedback Collection

Feedback collection involved gathering detailed input from users and stakeholders about their experiences using the prototype. User acceptance testing was conducted to assess ease of use and navigation, clarity of financial lessons, enjoyment and motivation levels, and appropriateness of gamified elements. This feedback provided valuable insight into user satisfaction and learning outcomes. It also guided us on the necessary adjustments to improve the system's usability, interactivity, and educational content.

3.8 Refinement

Based on the feedback obtained, the prototype was refined to address identified issues and enhance functionality. Changes were made to improve the visual design, simplify user interaction, and strengthen the educational content. Additional features that the users wanted to see there will be incorporated. The refined prototype was then re-tested with a larger group to ensure we met the expectations of users and be aligned with the project objectives.

3.9 Implementation and Maintenance

After successful testing and refinement, the final system was implemented and deployed for use. Maintenance activities were established to ensure continuous performance, data security, and timely updates. The system was designed to be scalable and adaptable, allowing future enhancements such as integration with educational institutions or financial literacy programs. Maintenance included regular updates to improve user experience, fixing bugs and errors, and monitoring user feedback for continuous improvement. This ensured that the platform remains relevant, reliable, and beneficial to its users over time.

3.10 Summary

This chapter outlined the prototyping methodology that was employed in the development of the interactive digital banking simulation platform. By following an iterative and user centered approach, the methodology ensured the creation of a robust, engaging, and educational system that effectively promotes financial literacy among children and teenagers in Uganda. Through continuous testing, feedback, and refinement, the final product was designed to provide an enjoyable and practical learning experience while supporting the broader goal of improving youth financial education in the country.

4 CHAPTER FOUR: SYSTEM DEVELOPMENT AND IMPLEMENTATION

4.1 Requirements Identified

4.1.1 User Requirements

User requirements for the Kids Banking System were identified through consultations with the key stakeholders of the system, namely parents or guardians who oversee financial activities, children who are the primary users of the platform for learning and managing money, and system administrators who ensure smooth operation, monitoring, and maintenance of the platform. These stakeholders highlighted several challenges associated with children's financial education and management. Parents expressed concerns about the lack of controlled environments where children can learn how to save, spend, and make financial decisions safely. Existing financial systems are often designed for adults, making them complex and unsuitable for children. Additionally,

there is limited visibility and control for parents over how children use money, and few tools that combine financial management with educational support. From the requirements gathering exercise, it was established that the system should support a structured and supervised financial ecosystem where children can perform basic banking activities such as saving money, setting goals, and requesting withdrawals, while all sensitive operations remain under parental control. The system should allow parents to create and manage child accounts, fund wallets, monitor transactions, and approve or reject withdrawal requests. Children should be able to interact with the system through a simple and engaging interface that allows them to view balances, track savings goals, receive allowances, and participate in financial learning activities. The system should also promote responsible financial behavior through budgeting tools, rewards, and educational content such as quizzes and tips. Furthermore, users emphasized the importance of simplicity, security, and accessibility. The system should be easy to use for children of different age groups while providing advanced control features for parents. It should also ensure secure authentication, protect sensitive data, and provide timely notifications for all important activities. Administrators required a management interface to monitor system usage, track transactions, manage users, and respond to reported issues. Based on the findings, the following user requirements were identified:

4.1.2 User Requirements List

1. The system should allow a parent or guardian to register and create an account securely.
2. The system should allow parents to create and manage one or more child accounts.
3. The system should allow children to securely access their accounts using simple authentication methods such as a PIN or biometric option.
4. The system should provide a digital wallet for each child to store and manage funds.
5. The system should allow parents to fund a child's account through supported payment methods.
6. The system should allow children to view their account balance and transaction history.
7. The system should allow children to set and track savings goals.
8. The system should allow children to request withdrawals subject to parental approval.

9. The system should allow parents to approve, reject, or limit withdrawal requests.
10. The system should support allowance scheduling and reward-based learning through chores or tasks.
11. The system should provide notifications and alerts for key activities such as deposits, withdrawals, and approvals.
12. The system should provide financial education content such as lessons, quizzes, and tips.
13. The system should maintain accurate transaction records and generate reports for parents.
14. The system should provide an administrative dashboard for monitoring system performance and user activity.
15. The system should ensure secure, reliable, and efficient operation with minimal delays during transactions.

4.2 System Requirements (New System)

The system requirements define the hardware and software environment needed for the digital banking simulation platform to operate effectively across mobile and web platforms. These requirements ensure that the system runs efficiently for both children and parents while supporting secure financial interactions and real-time processing.

Minimum Requirements

These are the minimum specifications required for the system to function properly.

Hardware

- Dual-core processor
- 4 GB RAM
- 20 GB available storage
- Smartphone (Android/iOS) or computer
- Wi-Fi or mobile data connectivity

Software

- Modern web browser such as Chrome, Edge, or Firefox (for web access)
- Operating system such as Windows, Linux, macOS, Android, or iOS
- Node.js runtime environment (for backend server)

Recommended Requirements

These specifications are recommended to achieve better performance, stability, and user experience.

Hardware

- Intel Core i5 processor or equivalent
- 8 GB RAM or higher
- SSD storage
- Smartphone with updated OS version
- Stable high-speed internet connection (Wi-Fi or 4G/5G)

Software

- Updated modern browser
- Latest stable Node.js version
- Properly configured development environment
- Updated mobile operating system (Android/iOS)

4.3 System Constraints

The main constraints affecting the design and implementation of the banking system are as follows:

1. Dependence on internet connectivity for real-time transactions and communication between users.
2. Reduced performance on low-end devices, especially for mobile users.
3. Security risks associated with handling sensitive financial data if not properly managed.
4. Limitations of SQLite for large-scale or high-volume transaction systems.
5. Variations in performance across different devices, browsers, and operating systems.

4.3.1 Assumptions

The design of the system was based on the following assumptions:

1. Users have access to a smartphone or computer with internet connectivity.
2. Users can install or access the application via mobile or web platforms.
3. Parents and children have basic digital literacy skills to use the system.
4. The backend server is properly configured and running during system use.
5. Users will follow system rules such as approval requirements for withdrawals.

4.4 Architectural Design of the New System

4.4.1 Description of the Designed System

The designed system, the Kids Banking System, is a secure, interactive digital platform developed to support financial management and financial literacy for children under

parental supervision. The system is designed to operate as a mobile and web-based application, enabling children to learn and practice money management in a controlled and educational environment while allowing parents to oversee and regulate all critical financial activities. The system enables a parent or guardian to create and manage child accounts, fund digital wallets, approve transactions, and monitor financial behavior. Children interact with the system through a simplified interface that allows them to view balances, set savings goals, request withdrawals, and participate in financial learning activities. Access to sensitive operations such as withdrawals, transfers, and external payments is strictly controlled through parental approval mechanisms such as one-time passwords (OTP) or approval PINs. The system addresses common challenges such as lack of financial literacy among children, absence of controlled spending environments, and limited parental visibility over children's financial activities. It integrates core banking-like functions with educational tools, including savings tracking, allowance management, budgeting support, and gamified learning modules. The platform also provides notifications, transaction history, reporting, and administrative monitoring in a unified system. At the implementation level, the system follows a client-server architecture. The frontend (mobile or web application) provides user interaction interfaces for both parents and children. The backend server, built using Node.js, handles business logic, authentication, transaction processing, and approval workflows. A database system stores user data, transaction records, savings goals, and audit logs. Integration with external services such as mobile money, SMS gateways, and email systems enables funding, notifications, and communication. A key control rule in the system design is that no withdrawal, external transfer, or sensitive financial operation can be completed without explicit parental approval, ensuring safety, accountability, and controlled financial learning.

The Authentication Module was implemented to manage user identity and regulate access to protected system resources. This module ensures that only authorized users are permitted to access system functionalities according to their assigned roles.

User authentication was implemented using JSON Web Tokens (JWT), which provide a secure mechanism for session management. During the login process, user credentials entered through the interface were validated against records stored in the relational database. Passwords were not stored in plain text; instead, they were securely hashed using the bcrypt algorithm before storage. During authentication, the entered password was verified against the stored hashed value.

Upon successful authentication, the system generated two tokens: an access token and a refresh token. The access token was used to authorize requests to protected endpoints, while the refresh token enabled renewal of user sessions without requiring repeated login. Role-based access control was also enforced, ensuring that system features were restricted according to user roles such as student, Academic Registrar staff, and administrator.

This module provided the security foundation for the entire system by ensuring confidentiality, maintaining session integrity, and controlling access to sensitive operations such as ticket handling, document management, and administrative functions.

4.5 Hardware Specifications

The system is designed to operate on commonly available consumer devices to ensure accessibility and affordability for families.

Component	Minimum Specification
Parent device	Smartphone or laptop with 2 GB RAM
Child device	Smartphone or tablet with basic app support
Network	Mobile data or Wi-Fi connection
Server host	Entry-level cloud server or local machine
Storage	At least 10 GB free storage

Table 5: System Hardware Requirements

Component	Minimum Specification
Parent device	Smartphone or laptop with 2 GB RAM
Child device	Smartphone or tablet with basic app support
Network	Mobile data or Wi-Fi connection
Server host	Entry-level cloud server or local machine
Storage	At least 10 GB free storage

Table 6: Recommended System Requirements

Component	Minimum Specification
Parent device	Smartphone or laptop with 4 GB RAM or higher
Child device	Tablet or smartphone with stable performance
Network	Stable 4G/5G or broadband internet
Server host	Cloud-based server
Storage	SSD storage for improved performance

Table 7: Minimum Specification

4.5.1 Minimum Hardware Specifications

4.5.2 Recommended Hardware Specifications

4.6 Software Specifications

4.7 Software Design Observations

- The backend server handles user authentication, wallet management, transaction processing, approval workflows, and reporting.
- The system uses secure APIs to communicate between frontend and backend components.
- OTP mechanisms are implemented for transaction approval and enhanced security.
- The database stores user accounts, transaction logs, savings goals, approvals, and audit records.
- External integrations (SMS, email, mobile money) are used to support notifications and financial transactions.
- The system is designed to be modular, allowing independent updates to components such as payments, notifications, and educational content.
- Security is prioritized through encryption, authentication controls, and monitoring mechanisms to protect sensitive financial data.

Software Component	Specification / Role
Operating system	Android, iOS, Windows, Linux, macOS
Runtime environment	Node.js
Backend framework	Express.js
Frontend technologies	HTML, CSS, JavaScript / Mobile frameworks
Database	SQLite / PostgreSQL
Authentication	JWT, OTP-based verification
Notification services	SMS gateway, Email services
Payment integration	Mobile money APIs, banking APIs
Development tools	VS Code, npm, Git
Security mechanisms	Encryption (AES, TLS), hashing
Browser support	Chrome, Edge, Firefox, mobile browsers

Table 8: Software Specifications

4.8 System Architecture

The Kids Banking System follows a layered client-server architecture designed to ensure secure financial transactions, real-time communication, and efficient

management of user data. The architecture separates responsibilities across different layers to improve scalability, maintainability, and security.

4.8.1 Architectural Layers

Presentation Layer

- Parent/Guardian mobile or web interface
- Child user interface (simplified dashboard)
- Administrator dashboard

This layer handles user interaction such as viewing balances, submitting withdrawal requests, setting savings goals, approving transactions, and accessing reports.

Application Layer

This layer contains the core business logic of the system.

- Node.js / Express backend server
- Authentication and authorization logic (role-based access control)
- Wallet and account management logic
- Transaction processing engine
- Savings goals and allowance management logic
- Approval workflow (OTP/PIN validation and processing)

This layer ensures that all financial operations are validated, authorized, and processed securely before any changes are made to the system.

Communication Layer

This layer handles interaction between the frontend, backend, and external services.

- RESTful APIs for client-server communication
- OTP delivery via SMS and email services
- Mobile money and payment gateway integrations
- Notification services (push notifications, email, SMS alerts)

This layer ensures that users receive real-time updates for deposits, withdrawals, approvals, and system alerts.

Data Layer

This layer is responsible for storage and data management.

- Relational database (SQLite or PostgreSQL)
- Tables for users, wallets, transactions, savings goals, approvals, and audit logs
- Secure storage of encrypted sensitive data
- Backup and recovery storage mechanisms

All financial records and user activities are persistently stored in this layer for accountability and reporting.

Critical Business Rules

- A child cannot withdraw funds directly without parental authorization.
- Every withdrawal request must trigger an approval PIN or code sent to the parent's verified phone or email.
- Approval codes must expire after a defined time.
- Withdrawal limits may vary by child age, account type, or parent settings.
- A parent may approve, reject, or ignore a request.
- Unapproved requests should expire automatically.

4.9 Suggested modules

A clean system design would include these modules: user and identity management, child account management, parent control panel, wallet and transactions, approval engine, notifications engine, savings and goals, education and gamification, reporting and analytics, security and audit logging, and integration services.

4.10 Tools and Technologies

The development of the banking system required the use of various software tools and technologies to support backend processing, database management, frontend interaction, and system testing. These tools were selected based on their reliability, scalability, ease of development, and ability to support secure financial operations. Each technology played a specific role in building an efficient and user-friendly banking system.

4.11 Programming Languages

Frontend Framework and Runtime

4.11.1 Expo with React Native

The frontend of the system was developed using Expo with React Native, which allows the application to run on multiple platforms including Android, iOS, and web. React Native was chosen because it enables the development of a single codebase that can be deployed across different devices, reducing development time and ensuring consistency in user experience. Expo simplifies the development process by providing built-in tools for building, testing, and deploying the application. Additionally, the system uses reactnative-web, which extends React Native functionality to web browsers, ensuring that users can access the banking system from mobile devices and desktop browsers seamlessly.

4.11.2 Backend Language and Framework

The backend of the system was developed using TypeScript running on Node.js, which provides a strongly typed and structured environment for server-side development. TypeScript improves code reliability by reducing runtime errors and making the system easier to maintain. The backend framework used is Express, which is a lightweight and

flexible Node.js framework for building RESTful APIs. Express was used to handle core banking operations such as user registration, login, transaction processing, savings management, and approval workflows between parents and children. It acts as the central communication layer between the frontend and the database.

4.11.3 API Validation and Security

To ensure that all incoming data is correct and secure, the system uses Zod for API validation. Zod is a schema validation library that checks the structure and type of data before it is processed by the backend, preventing invalid or malicious inputs from affecting the system. For security, the system uses bcryptjs to hash user passwords, ensuring that sensitive credentials are not stored in plain text. Authentication is managed using JSON Web Tokens (JWT), which allow secure session handling between the user and the system. In addition, cookie-parser and CORS (Cross-Origin Resource Sharing) are used to manage secure session cookies and control access between the frontend and backend.

4.11.4 API Validation and Security

To ensure that all incoming data is correct and secure, the system uses Zod for API validation. Zod is a schema validation library that checks the structure and type of data before it is processed by the backend, preventing invalid or malicious inputs from affecting the system. For security, the system uses bcryptjs to hash user passwords, ensuring that sensitive credentials are not stored in plain text. Authentication is managed using JSON Web Tokens (JWT), which allow secure session handling between the user and the system. In addition, cookie-parser and CORS (Cross-Origin Resource Sharing) are used to manage secure session cookies and control access between the frontend and backend.

4.11.5 Database and ORM

The system uses Prisma as the Object Relational Mapping (ORM) tool to manage database interactions. Prisma allows developers to interact with the database using TypeScript instead of writing raw SQL queries, making the system more structured and easier to maintain. It is responsible for handling operations such as creating users, updating account balances, storing transactions, and managing savings goals. The database engine used in the system is SQLite, which provides a lightweight and file-based relational database solution suitable for development and small to medium-scale applications. The database structure is defined in the schema.prisma file, which ensures clear data modeling and consistency across the system.

4.12 Business Rules

BR1: No withdrawal without parent approval All withdrawal requests initiated by a child must be approved by the parent or guardian before the transaction is processed. This rule ensures that all financial activities remain under adult supervision for security and control.

BR2: OTP expires after defined duration The one-time password (OTP) generated for transaction approval is only valid for a limited period. This reduces the risk of unauthorized access and ensures timely validation of transactions.

BR3: Child cannot access parent controls Children are restricted from accessing features reserved for parents, such as approval of transactions and account settings.

This separation of roles ensures system security and proper control.

BR4: Maximum withdrawal limits configurable Parents can set limits on the amount a child can withdraw within a specific period. This helps manage spending behavior and prevents excessive withdrawals.

4.13 Risks & Mitigation

OTP delays → Fallback channels In case of delays in OTP delivery, alternative communication channels such as email can be used to ensure timely transaction approval and avoid user inconvenience.

Fraud attempts → Lock accounts If suspicious activities or multiple failed authentication attempts are detected, the system will temporarily lock the account to prevent unauthorized access and protect user funds.

System downtime → Redundancy To minimize service interruptions, the system implements backup and redundancy mechanisms. This ensures continued operation even in the event of system failures.

4.14 Future Enhancements

AI financial advisor The system may incorporate an AI-based assistant to provide personalized financial advice and recommendations to children based on their spending and saving patterns. **Integration with schools** The platform can be integrated with educational institutions to support financial literacy programs and track student learning progress. **Debit card issuance** Future versions of the system may include issuing controlled debit cards for children, allowing them to make real-world

transactions under parental supervision. Gamified leaderboards The system may introduce leaderboards and rewards to encourage competition and engagement among users, making financial learning more interactive and enjoyable.

4.15 System Flow

The system flow of the Kids Banking System describes how users (Parents/Guardians, Children, and Administrators) interact with the system from account access through financial operations, learning activities, and reporting. The flow ensures that all financial actions are properly authorized, recorded, and monitored within a secure and controlled environment.

Parent/Guardian Flow:

- The parent logs in using secure authentication (password + OTP).
- The parent accesses the dashboard showing child accounts and financial overview.
- The parent can create or manage child profiles.
- The parent funds the child wallet using supported payment methods.
- The parent receives and reviews withdrawal requests from children.
- The parent approves or rejects transactions using OTP/PIN verification.

Child Flow:

- The child logs in using PIN, biometric, or device authentication.
- The system displays the child dashboard (balance, savings goals, allowance, and rewards).
- The child can set or update savings goals.
- The child can view deposits and transaction history.
- The child submits withdrawal requests when needed.

- The child participates in financial education activities (quizzes, lessons, and challenges).

Transaction Processing Flow:

- The system validates all requests (deposit, withdrawal, or transfer).
- Withdrawal or external payment requests are forwarded to the parent for approval.
- The parent receives an OTP or approval notification.
- Upon approval, the system processes the transaction and updates the wallet balance.
- The system logs all financial activities for auditing purposes.

Notification Flow:

- The system sends real-time notifications for deposits, withdrawals, approvals, rejections, and savings milestones.
- Notifications are delivered via SMS, email, or in-app alerts.

Reporting and Monitoring Flow:

- The system records all transactions, approvals, and user activities.
- Parents can view reports on spending patterns, savings progress, and financial behavior.
- Administrators access system dashboards for user management, analytics, and audit logs.

The following diagrams were developed during the quick design phase:

- **Use Case Diagram:** Illustrated interactions between system users (child, parent, and system administrator) and system functionalities.
- **Entity Relationship Diagram (ERD):** Models the database structure and relationships between entities such as users, accounts, transactions, and savings goals, showing how data is stored, linked, and managed within the banking system.
- **Class Diagram:** Illustrates system object structures, attributes, and relationships among classes.

These diagrams ensured clarity in the banking system's operations, improved communication of design decisions such as account management, transaction processing, and parental controls, and guided the implementation process. The structured modeling approach helped visualize how users, accounts, transactions, and savings goals interact, ensuring the system was well-designed and aligned with software engineering best practices before development began.

4.16 Use Case Diagram (Textual)

This section illustrates the interactions between the different actors in the system, namely the Child, Parent/Guardian, and Administrator, and the various functionalities they can perform. It provides a high-level view of the system by showing how users engage with features such as account management, transaction requests, approvals, and monitoring activities.

Actors and Interactions

- **Child**
 - Request Withdrawal
 - View Balance – Manage Savings Goals
- **Parent/Guardian**
 - Approve Transactions
 - Fund Child Account
 - Set Parental Controls

- **Administrator**

- Monitor System Activities

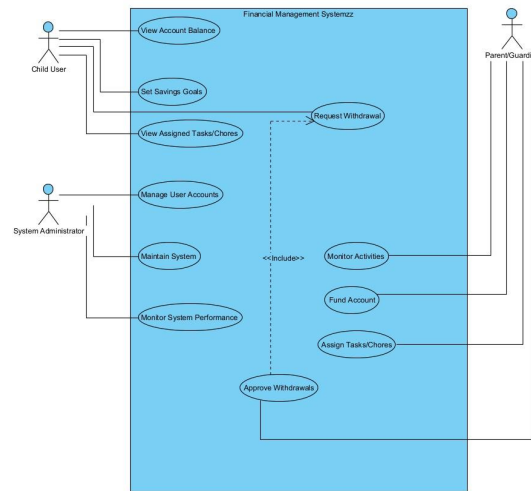


Figure 4: usecase

This use case diagram represents a financial literacy system designed to balance a child’s independence with structured parental oversight. In this system, the Child User and Parent/Guardian act as the primary actors whose interactions form a continuous feedback loop. The child engages with the platform by managing tasks such as completing chores, setting savings goals, and requesting withdrawals, which helps them develop practical financial skills. At the same time, the parent supports and regulates this learning process by funding the child’s account, assigning tasks, and approving or rejecting financial requests. This arrangement should ensure that the child gains hands-on experience in managing money while still operating within a controlled and guided environment. The System Administrator serves as a secondary actor responsible for maintaining the technical stability and security of the platform. Their role focuses on backend operations such as managing user accounts, monitoring system performance, and ensuring the system functions reliably. Unlike the parent and child, the administrator does not participate in daily financial activities but instead provides the necessary support to keep the system secure and efficient. This clear separation of responsibilities enhances the overall design by allowing family interactions to remain the focus, while the administrator ensures a safe and dependable environment for financial learning.

4.17 Sequence Diagram

The sequence diagram for the withdrawal process illustrates the interaction between the child, system, OTP service, and parent to ensure secure transaction execution. It begins when the child initiates a withdrawal request, after which the system generates a

one-time password (OTP) and sends it to the parent for authorization. The parent then submits the OTP back to the system, where it is validated to confirm its correctness and validity. Once successfully verified, the system proceeds to deduct the requested amount from the child's wallet and updates the account balance. Finally, notifications are sent to both the child and the parent to inform them of the transaction outcome, ensuring transparency and maintaining a secure, parent-controlled withdrawal process.

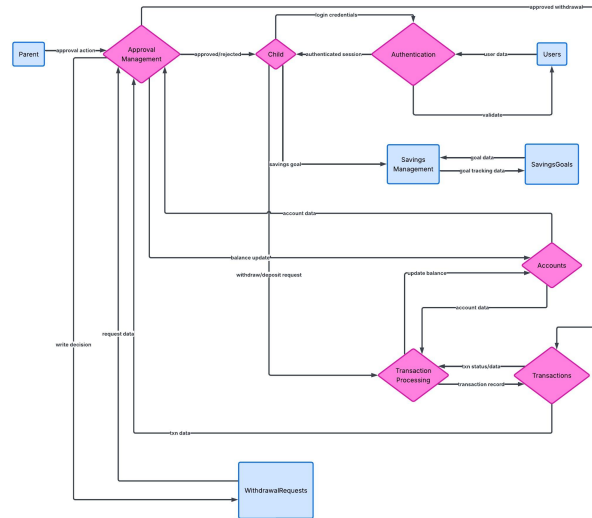


Figure 5: sequence diagram

4.18 Class Diagram

This represents the structural design of the system by showing the main classes, their attributes, and the relationships between them. It illustrates how different components such as users, accounts, transactions, and savings goals are organized and connected within the system. This diagram is important for understanding how data is structured and how objects interact during system implementation.

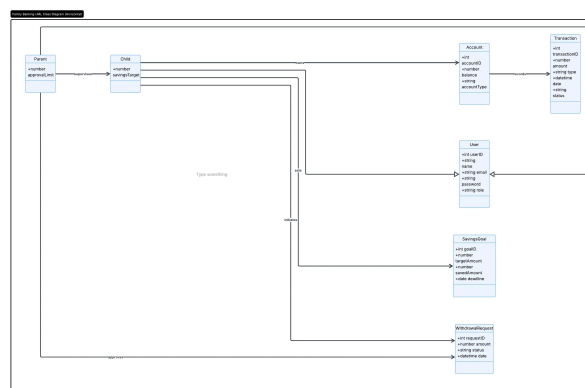


Figure 6: Classdiagram

4.19 Flow Chart

The flowchart shows the step-by-step flow of processes within the system, particularly how tasks and decisions are carried out. It uses symbols and arrows to represent actions, conditions, and the sequence of operations, such as the withdrawal approval process. This diagram helps to clearly visualize the system logic and makes it easier to understand how different processes are executed from start to finish.

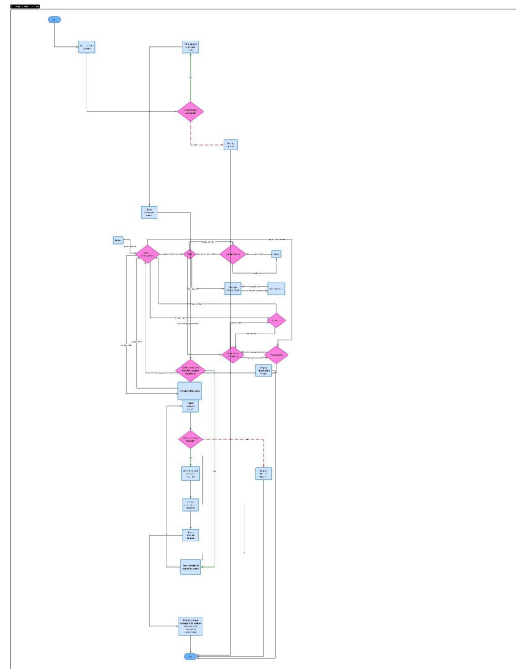


Figure 7: flowchart

4.20 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) shows how data moves through your banking system, from the users to the system processes and into the database. It illustrates how information such as login details, transaction requests, and approval decisions is received, processed, stored, and returned as output. In your system, the DFD helps visualize how the child and parent interact with different system functions like authentication, transaction processing, and approval management, while also showing how all data is stored and retrieved from the database. It focuses on the flow of data rather than the step-by-step logic of operations.

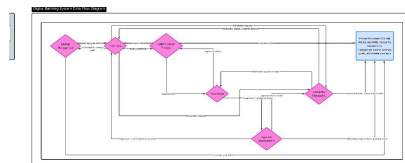


Figure 8: dataflow diagram

4.21 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the structure of the Kids Banking System and the relationships between its core entities. The main entities include Parent, Child, Wallet, Transaction, Withdrawal Request, Approval, Savings Goal, Allowance, Notification, and Audit Log.

A Parent can manage multiple Child accounts, forming a one-to-many relationship. Each child is linked to a single Wallet, which stores their balance and supports financial activities. The Transaction entity records all deposits, withdrawals, and transfers associated with a wallet.

Children can initiate Withdrawal Requests, which must be authorized through the Approval entity by a parent, ensuring secure and controlled financial operations. The system also includes Savings Goals and Allowance features to support financial learning and money management.

Additionally, Notifications keep users informed about system activities, while Audit Logs maintain records of all actions for security and accountability. Overall, the ERD provides a clear representation of how data is organized and how entities interact within the system.

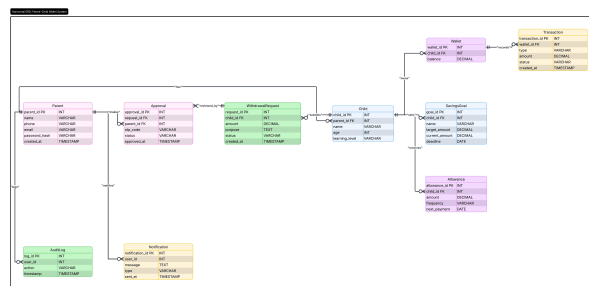


Figure 9: ER Diagram

4.22 SYSTEM FORMS

4.22.1 PARENT DASHBOARD

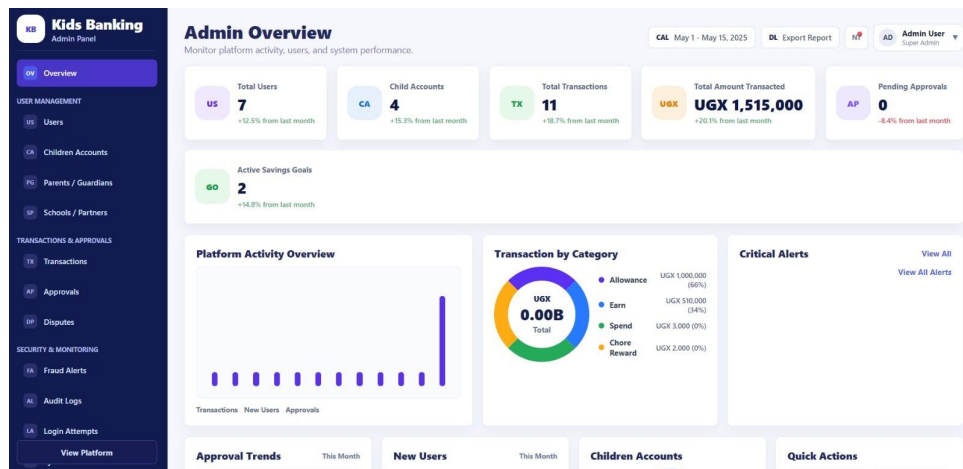


Figure 10: parentdashboard

4.22.2 Child dashboard

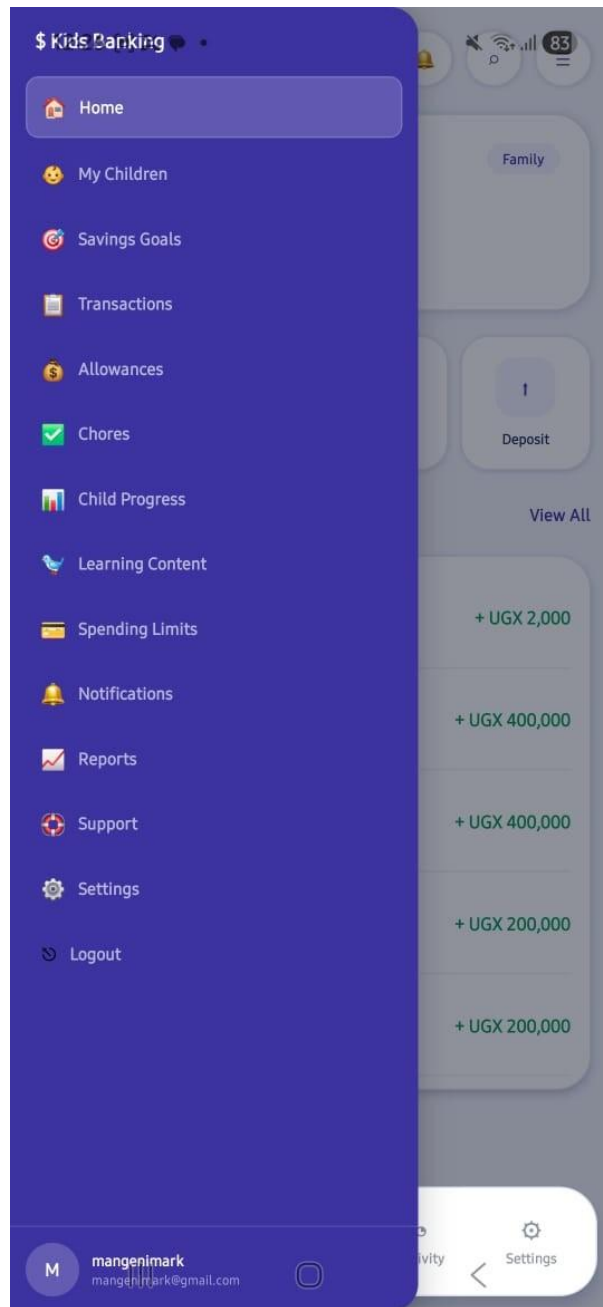


Figure 11: Child dashboard

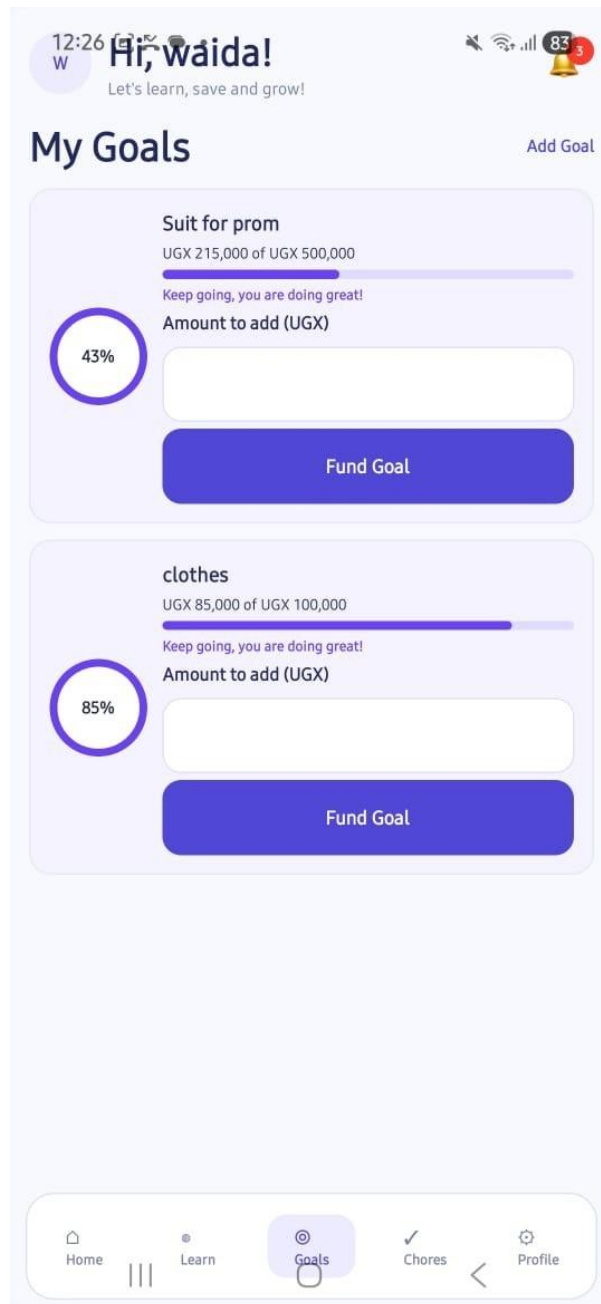


Figure 12: Child mobile page

5 CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents a summary of the study, conclusions drawn from the development and testing of the digital banking simulation platform, and recommendations for its

adoption and future improvement. It also outlines areas for further enhancement based on system limitations identified during implementation and evaluation.

5.2 Summary

The digital banking simulation system was developed as an interactive platform aimed at improving financial literacy among teenagers through practical engagement with budgeting, saving, and responsible spending. The system was designed to address challenges such as lack of financial education tools, limited practical exposure to money management, and the absence of controlled environments where young users can safely learn financial decision-making. The system provides core functionalities such as virtual wallet management, savings goal tracking, budgeting tools, transaction history, and a parental approval mechanism for withdrawals. It also includes authentication features, secure transaction handling, and a structured database system to ensure accurate financial record management. The platform was implemented using a cross-platform mobile and web architecture, enabling accessibility on Android, iOS, and web devices. From a personal perspective, the implementation of this system demonstrates that financial literacy can be effectively taught through interactive digital environments. By simulating real-world banking operations within a controlled system, users are able to practice financial decision-making without real financial risk. This aligns with studies that emphasize the importance of experiential learning in improving financial behavior among young users. From a broader perspective, the system contributes to the growing need for digital financial education tools that support early financial awareness. It provides an accessible and structured environment where teenagers can learn essential money management skills, while parents maintain oversight and guidance through approval-based controls.

5.3 Conclusion

5.3.1 Legal and Ethical Considerations

The development and use of the banking simulation system raise important ethical considerations, particularly regarding data privacy and responsible usage. The system handles sensitive user information such as account details, transaction history, and savings records; therefore, appropriate security measures such as password encryption and secure authentication mechanisms were implemented to protect user data. Ethically, the system ensures that teenagers are guided rather than given unrestricted financial control. Parental approval mechanisms were included to promote responsible financial behavior while maintaining user safety. Users must also be informed about data usage within the system to ensure transparency and trust.

5.3.2 Positive Implications of the System

The system provides several benefits to both teenagers and parents. It enhances financial literacy by allowing users to learn budgeting, saving, and spending through real-life simulations. The inclusion of interactive features such as savings goals and transaction tracking helps users develop practical financial habits. For parents, the system offers oversight tools that allow them to monitor financial activities and approve withdrawals, ensuring controlled financial learning. At a broader level, the system contributes to bridging the gap in financial education by providing a digital, accessible, and engaging learning platform.

5.3.3 Achievement of Objectives

The project successfully achieved its main objectives. The system was able to design and implement an interactive financial simulation platform that supports budgeting, savings management, and controlled spending for teenagers. It also successfully integrated a parental approval system to ensure guided financial learning. In addition, the system met key functional and non-functional requirements such as usability, responsiveness, security, and cross-platform accessibility. However, further improvements are required in areas such as scalability testing and integration with real banking systems for broader applicability.

5.3.4 Factors Influencing Adoption

Several factors influence the adoption of the system, including user familiarity with digital platforms, ease of use, and perceived usefulness in improving financial skills. The success of the system also depends on parental engagement and willingness to participate in monitoring and approval processes. Technical factors such as device compatibility, internet access, and system performance also affect usability. Additionally, user acceptance is influenced by how well the system demonstrates value in teaching real-world financial skills in an engaging and simple manner.

5.3.5 Recommendations

The system is recommended for adoption in educational environments and homes as a financial literacy tool for teenagers. It should be introduced through pilot testing before full deployment to ensure usability and effectiveness across different user groups. It is recommended that the system be enhanced with additional features such as AI-based financial advice, gamification improvements, and personalized budgeting recommendations to further increase user engagement. Strong security enhancements such as multi-factor authentication should also be implemented to improve system

safety. Institutions and parents should provide guidance and supervision during initial usage to ensure proper understanding of financial concepts. Continuous system updates based on user feedback are also recommended to improve performance and usability.

5.3.6 Areas for Further Study

Although the system achieved its primary objectives, further research is needed to improve its scalability and functionality. Future enhancements may focus on integrating real banking APIs to simulate real-world financial transactions more accurately. Additional improvements could include the development of advanced analytics for spending behavior, AI-driven financial coaching, and offline functionality to improve accessibility. Further studies should also explore long-term impacts of digital financial simulation tools on teenage financial behavior and decision-making skills.

5.4 APPENDICES

5.4.1 End User Guide

Introduction

This appendix provides a practical guide for using the Kids Banking System. It explains how to register, log in, manage child accounts, perform financial activities such as deposits and withdrawals, approve transactions, and access administrative features.

The Kids Banking System is designed as a secure digital financial platform that enables children to learn money management under parental supervision. The system supports both mobile and web access, allowing users to perform financial operations, track savings goals, and receive real-time notifications.

Who This Guide Is For

- **Parent/Guardian** – Manages child accounts, funds wallets, approves transactions, and monitors financial activity.
- **Child User** – Views balance, sets savings goals, requests withdrawals, and participates in learning activities.
- **System Administrator** – Monitors system performance, manages users, and reviews transactions and reports.

Before You Start

1. A smartphone, tablet, or computer is available.
2. Internet connection (mobile data or Wi-Fi) is active.
3. A modern browser or mobile app is installed.
4. The system URL or application is accessible.
5. The parent has a valid phone number/email for OTP verification.

Starting the System

Step 1: Open the Application

- Mobile application, or
- Web browser (e.g., <http://localhost:3000> or deployed URL)

Step 2: Register or Log In

- New users select **Register**
- Existing users select **Login**

Step 3: Authentication

- Parent logs in using email/phone + password + OTP
- Child logs in using PIN or biometric authentication

How to Use the System as a Parent/Guardian

1. Access dashboard (child accounts, balances, approvals, transactions)
2. Create child account (name, age, learning level, savings goals)

3. Fund wallet (enter amount, select payment method, confirm)
4. Manage withdrawal requests (approve/reject via OTP/PIN)
5. Set controls (limits, spending caps, freeze/unfreeze)
6. Monitor activity (history, savings progress, reports)

How to Use the System as a Child

1. Log in using PIN or biometric authentication
2. View dashboard (balance, goals, rewards)
3. Create savings goals (name, amount, deadline)
4. Request withdrawal (enter amount and submit)
5. View transactions and track progress
6. Participate in learning activities (lessons, quizzes)

Transaction Flow

1. Parent funds wallet or child initiates request
2. System validates request
3. Parent receives OTP for approval
4. Parent approves/rejects
5. System updates balance
6. Transaction is logged

Notifications and Alerts

- Deposits

- Withdrawal requests
- Approvals/rejections
- Low balance
- Savings milestones

Delivery Channels:

- SMS
- Email
- In-app notifications

Admin Dashboard Usage

1. Open dashboard (admin URL)
2. Log in (username and password)
3. View users, transactions, reports, logs
4. Manage system performance and issues

Reports and Data Export

- Transaction history
- Savings progress
- User activity

Common Buttons and Their Meaning

Button	Function
Register	Create new account

Login	Access system
Fund Wallet	Add money to child account
Request Withdrawal	Initiate withdrawal
Approve	Confirm transaction
Reject	Decline transaction
Create Goal	Add savings goal
View Reports	Open analytics dashboard
Logout	Exit system

Troubleshooting Guide

- **Cannot Log In:** Verify credentials or request new OTP.
- **Transaction Not Processed:** Wait for parent approval.
- **OTP Not Received:** Retry or check phone/email.
- **Incorrect Balance:** Refresh or contact support.

Recommended Good Practices

- Parents should regularly monitor transactions
- Children should set realistic savings goals
- Use strong passwords and secure PINs
- Do not share login credentials
- Review reports frequently

Shutting Down the System

1. Complete pending transactions
2. Log out of the system
3. Close the application

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